Task 1: Network Quality Dataset

Aim

To investigate whether high network traffic is associated with increased latency using the Network Quality Dataset.

Tool Used

Python / Google Colab. Python was used to analyze the dataset, calculate total traffic, group traffic levels, calculate average latency, and create the traffic-versus-latency graph.

Procedure

1. Loaded the Network Quality Dataset.

2. Identified the traffic_in_bps, traffic_out_bps, latency_ms, and timestamp columns.

3. Calculated total traffic as traffic_in_bps + traffic_out_bps.

4. Grouped the data into different traffic levels.

5. Calculated the average latency for each traffic level.

6. Created a graph comparing traffic level with average latency.

7. Observed periods where latency became noticeably high.

8. Suggested a network improvement based on the observation.

Graph

The graph below shows the relationship between total network traffic and average latency.

Observation

The average latency was approximately 95–98 ms for most traffic levels. At very high traffic levels above 10 Mbps, the average latency increased to approximately 102.48 ms.

The dataset also contains individual periods of noticeably high latency, including values of 175 ms and above. Therefore, high traffic may contribute to increased latency, although traffic is not the only factor affecting network performance.

Conclusion

Network latency is generally stable under normal traffic conditions but increases slightly at very high traffic levels. The analysis indicates that high traffic can contribute to increased latency, while other network factors may also affect latency.

Recommendation

The organization should increase network bandwidth/capacity at heavily loaded network points. This can reduce congestion during periods of very high traffic and improve network response time.

Average Latency by Traffic Level

Total Traffic Level	Average Latency (ms)
< 1 Mbps	98.21
1–2 Mbps	96.05
2–3 Mbps	95.63
3–4 Mbps	95.53
4–5 Mbps	95.23
5–10 Mbps	97.25
> 10 Mbps	102.48

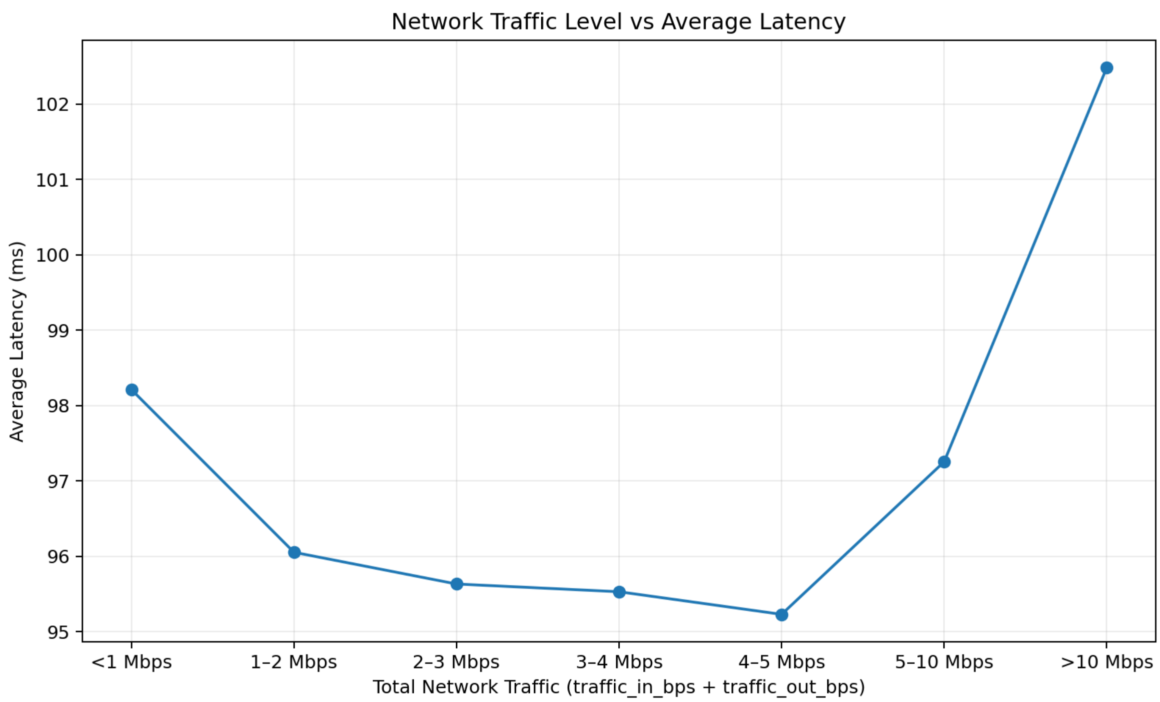

Figure 1: Network Traffic Level vs Average Latency